

Case Study No.: 1

Case Study Title

Student Information Portal with Welcome Alert using HTML and JavaScript

Theory

A **Student Information Portal** is a simple web application developed using **HTML** and **JavaScript** to display student details and interact with users through alert messages. JavaScript is used to make web pages interactive by responding to user actions such as button clicks.

In this case study, three methods of JavaScript implementation are demonstrated:

1. Inline JavaScript

Inline JavaScript is written directly inside an HTML element using event attributes such as **onclick**. It is mainly used for simple tasks like displaying an alert message.

Example:

```
<button onclick="alert('Welcome to SIT')">Click Here</button>
```

2. Internal JavaScript

Internal JavaScript is written inside the `<script>` tag within the HTML document. It is suitable for small web applications where all JavaScript code is contained in a single file.

Example:

- `function welcomeMessage(){`
- `alert("Welcome to Symbiosis Institute of Technology");`
- `}`

3. External JavaScript

External JavaScript is written in a separate `.js` file and linked to the HTML page using the `<script src="">` tag. This approach improves code organization, readability, and reusability.

Example:

```
<script src="script.js"></script>
```

Alert Box

An **Alert Box** is a JavaScript dialog box used to display important messages or notifications to the user. It appears on the screen until the user clicks the **OK** button.

Syntax:

```
alert("Welcome to SIT");
```

Console Methods

JavaScript provides various console methods for debugging programs.

- **console.log()** – Displays general information.
 - **console.info()** – Displays informational messages.
 - **console.warn()** – Displays warning messages.
 - **console.error()** – Displays error messages.
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Features of the Case Study

- Displays student information.
- Shows a welcome message using an alert box.
- Demonstrates Inline, Internal, and External JavaScript.
- Uses JavaScript event handling.
- Demonstrates console methods for debugging.
- Provides a simple and interactive user interface.

1. Case Study Program Code:

index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <meta name="description" content="SIT Nagpur - Department of Computer
Science & Engineering. Explore our programs, faculty, and student
resources." />
  <title>SIT Nagpur | Department of Computer Science</title>
  <link rel="preconnect" href="https://fonts.googleapis.com" />
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin />
```

```
<link
href="https://fonts.googleapis.com/css2?family=Outfit:wght@300;400;500;600;700;800;900&family=Inter:wght@300;400;500;600&display=swap"
rel="stylesheet" />
<style>
  *, *::before, *::after { margin: 0; padding: 0; box-sizing:
border-box; }

  :root {
    --primary:      #2563eb;
    --primary-dark: #1d4ed8;
    --accent:       #0ea5e9;
    --gold:         #d97706;
    --bg-dark:      #ffffff;
    --bg-card:      #ffffff;
    --bg-card2:     #f8fafc;
    --text-main:    #0f172a;
    --text-muted:   #475569;
    --border:       #e2e8f0;
    --shadow:       0 8px 40px rgba(0,0,0,0.05);
    --radius:       16px;
    --transition:   0.35s cubic-bezier(0.4,0,0.2,1);
  }

  html { scroll-behavior: smooth; }

  body {
    font-family: 'Inter', sans-serif;
    background: var(--bg-dark);
    color: var(--text-main);
    overflow-x: hidden;
  }

  body::before {
    content: '';
    position: fixed;
    inset: 0;
    background:
      radial-gradient(ellipse 80% 60% at 20% 10%, rgba(59,130,246,0.18)
0%, transparent 60%),
```

```

        radial-gradient(ellipse 60% 50% at 80% 80%, rgba(56,189,248,0.12)
0%, transparent 60%),
        radial-gradient(ellipse 50% 40% at 50% 50%, rgba(244,63,94,0.07)
0%, transparent 70%);
    pointer-events: none;
    z-index: 0;
}

/* NAVBAR */
nav {
    position: sticky; top: 0; z-index: 100;
    backdrop-filter: blur(20px);
    background: rgba(255,255,255,0.9);
    border-bottom: 1px solid var(--border);
    padding: 0 5%;
    display: flex; align-items: center; justify-content: space-between;
    height: 70px;
    animation: slideDown 0.6s ease;
}

@keyframes slideDown {
    from { transform: translateY(-100%); opacity: 0; }
    to   { transform: translateY(0);    opacity: 1; }
}

.nav-logo {
    display: flex; align-items: center; gap: 12px;
    text-decoration: none;
}

.logo-icon {
    width: 42px; height: 42px;
    background: linear-gradient(135deg, var(--primary), var(--accent));
    border-radius: 10px;
    display: grid; place-items: center;
    font-size: 20px; font-family: 'Outfit', sans-serif; font-weight:
800; color: #fff;
    box-shadow: 0 4px 18px rgba(59,130,246,0.5);
}

```

```
.nav-logo span {
  font-family: 'Outfit', sans-serif; font-weight: 700; font-size:
1.1rem;
}

.nav-links { display: flex; gap: 6px; list-style: none; }

.nav-links a {
  color: var(--text-muted); text-decoration: none;
  font-size: 0.9rem; font-weight: 500;
  padding: 8px 16px; border-radius: 8px;
  transition: var(--transition);
}

.nav-links a:hover { color: var(--text-main); background:
rgba(59,130,246,0.15); }
.nav-links a.active { color: var(--primary); background:
rgba(59,130,246,0.12); }

.nav-cta {
  background: linear-gradient(135deg, var(--primary),
var(--primary-dark)) !important;
  color: #fff !important;
  border-radius: 10px !important;
  box-shadow: 0 4px 16px rgba(59,130,246,0.4);
}

/* HERO */
.hero {
  position: relative; z-index: 1;
  min-height: calc(100vh - 70px);
  display: flex; flex-direction: column; align-items: center;
justify-content: center;
  text-align: center; padding: 80px 5% 60px;
}

.hero-badge {
  display: inline-flex; align-items: center; gap: 8px;
  background: rgba(59,130,246,0.15);
  border: 1px solid rgba(59,130,246,0.35);
```

```
border-radius: 50px; padding: 8px 20px;
font-size: 0.82rem; font-weight: 600;
color: var(--accent); letter-spacing: 0.06em; text-transform:
uppercase;
margin-bottom: 28px;
animation: fadeUp 0.8s ease 0.2s both;
}

.hero-badge::before {
  content: '●'; font-size: 8px; color: var(--accent);
  animation: pulse 2s infinite;
}

@keyframes pulse { 0%,100%{opacity:1} 50%{opacity:0.3} }

.hero h1 {
  font-family: 'Outfit', sans-serif;
  font-size: clamp(2.4rem, 6vw, 4.8rem);
  font-weight: 900; line-height: 1.1;
  margin-bottom: 24px;
  animation: fadeUp 0.8s ease 0.35s both;
}

.highlight {
  color: var(--primary);
}

.hero p {
  max-width: 620px; color: var(--text-muted);
  font-size: 1.08rem; line-height: 1.8;
  margin-bottom: 44px;
  animation: fadeUp 0.8s ease 0.5s both;
}

.hero-btns {
  display: flex; gap: 16px; flex-wrap: wrap; justify-content: center;
  animation: fadeUp 0.8s ease 0.65s both;
}

@keyframes fadeUp {
```

```

    from { opacity:0; transform:translateY(30px); }
    to   { opacity:1; transform:translateY(0); }
  }

  .btn-primary {
    background: linear-gradient(135deg, var(--primary),
var(--primary-dark));
    color: #fff; border: none;
    padding: 16px 36px; border-radius: 14px;
    font-size: 1rem; font-weight: 600; cursor: pointer;
    text-decoration: none; display: inline-flex; align-items: center;
gap: 10px;
    box-shadow: 0 6px 28px rgba(59,130,246,0.45);
    transition: var(--transition); font-family: 'Inter', sans-serif;
  }

  .btn-primary:hover { transform: translateY(-3px); box-shadow: 0 12px
40px rgba(59,130,246,0.6); }

  .btn-secondary {
    background: rgba(0,0,0,0.03);
    color: var(--text-main);
    border: 1px solid var(--border);
    padding: 16px 36px; border-radius: 14px;
    font-size: 1rem; font-weight: 600; cursor: pointer;
    text-decoration: none; display: inline-flex; align-items: center;
gap: 10px;
    backdrop-filter: blur(10px);
    transition: var(--transition); font-family: 'Inter', sans-serif;
  }

  .btn-secondary:hover { background: rgba(59,130,246,0.12);
border-color: var(--primary); transform: translateY(-3px); }

/* STATS */
  .stats-strip {
    position: relative; z-index: 1;
    display: flex; justify-content: center; flex-wrap: wrap;
    background: rgba(248,250,252,0.8);
    border-top: 1px solid var(--border);

```



```
border-bottom: 1px solid var(--border);
backdrop-filter: blur(12px);
}

.stat-item {
  flex: 1; min-width: 160px;
  padding: 32px 20px; text-align: center;
  border-right: 1px solid var(--border);
  transition: var(--transition);
}

.stat-item:last-child { border-right: none; }
.stat-item:hover { background: rgba(59,130,246,0.07); }

.stat-num {
  font-family: 'Outfit', sans-serif;
  font-size: 2.4rem; font-weight: 800;
  color: var(--primary);
  display: block;
}

.stat-label { font-size: 0.85rem; color: var(--text-muted);
font-weight: 500; margin-top: 4px; }

/* SECTION COMMON */
section { position: relative; z-index: 1; padding: 90px 5%; }

.section-tag {
  display: inline-block;
  background: rgba(59,130,246,0.12);
  border: 1px solid rgba(59,130,246,0.3);
  color: var(--primary);
  font-size: 0.78rem; font-weight: 700;
  letter-spacing: 0.1em; text-transform: uppercase;
  padding: 6px 16px; border-radius: 50px; margin-bottom: 18px;
}

.section-title {
  font-family: 'Outfit', sans-serif;
  font-size: clamp(1.8rem, 3.5vw, 2.8rem);
```

```
    font-weight: 800; margin-bottom: 14px;
  }

  .section-sub {
    color: var(--text-muted); font-size: 1rem;
    line-height: 1.7; max-width: 560px; margin-bottom: 56px;
  }

  /* ABOUT */
  #about {
    display: grid; grid-template-columns: 1fr 1fr;
    gap: 64px; align-items: center;
  }

  .about-visual { position: relative; }

  .about-card-main {
    background: linear-gradient(135deg, var(--bg-card),
var(--bg-card2));
    border: 1px solid var(--border);
    border-radius: 24px; padding: 40px;
    box-shadow: var(--shadow);
  }

  .about-icon {
    width: 60px; height: 60px;
    background: linear-gradient(135deg, var(--primary), var(--accent));
    border-radius: 14px;
    display: grid; place-items: center;
    font-size: 28px; margin-bottom: 20px;
    box-shadow: 0 8px 24px rgba(59,130,246,0.4);
  }

  .about-card-main h3 { font-family: 'Outfit', sans-serif; font-size:
1.4rem; font-weight: 700; margin-bottom: 12px; }
  .about-card-main p { color: var(--text-muted); line-height: 1.7;
font-size: 0.95rem; }

  .floating-badge {
    position: absolute;
```

```

    background: linear-gradient(135deg, var(--gold), #f59e0b);
    color: #1a1400; font-weight: 700; font-size: 0.82rem;
    padding: 10px 18px; border-radius: 50px;
    box-shadow: 0 8px 24px rgba(251,191,36,0.4);
}

.floating-badge.top-right { top: -18px; right: -18px; }
.floating-badge.bot-left { bottom: -18px; left: -18px; }

.feature-list { list-style: none; display: flex; flex-direction:
column; gap: 14px; }

.feature-list li {
    display: flex; align-items: center; gap: 12px;
    font-size: 0.95rem; color: var(--text-muted);
}

.feature-list li::before {
    content: '✓';
    width: 26px; height: 26px;
    background: rgba(56,189,248,0.15);
    border: 1px solid rgba(56,189,248,0.35);
    color: var(--accent); border-radius: 50%;
    display: grid; place-items: center;
    font-size: 0.7rem; font-weight: 800; flex-shrink: 0;
}

/* PROGRAMS */
#programs { background: #f8fafc; }

.programs-list {
    display: flex;
    flex-direction: column;
    gap: 16px;
    max-width: 1000px;
    margin: 0 auto;
}

.program-row {
    display: grid;

```

```
    grid-template-columns: auto 1fr auto;
    gap: 40px;
    align-items: center;
    padding: 32px 40px;
    background: var(--bg-card);
    border: 1px solid var(--border);
    border-radius: 16px;
    transition: var(--transition);
    cursor: pointer;
}

.program-row:hover {
    border-color: var(--primary);
    box-shadow: 0 12px 30px rgba(0,0,0,0.04);
    transform: translateX(4px);
}

.prog-number {
    font-family: 'Outfit', sans-serif;
    font-size: 2.5rem;
    font-weight: 800;
    color: var(--border);
    transition: var(--transition);
}

.program-row:hover .prog-number {
    color: var(--primary);
}

.prog-content h3 {
    font-family: 'Outfit', sans-serif;
    font-size: 1.3rem;
    font-weight: 700;
    margin-bottom: 8px;
    color: var(--text-main);
}

.prog-content p {
    color: var(--text-muted);
    font-size: 0.95rem;
```

```
    line-height: 1.6;
    max-width: 600px;
}

.prog-meta {
    text-align: right;
}

.prog-duration {
    display: block;
    color: var(--primary);
    font-size: 0.95rem;
    font-weight: 700;
    margin-bottom: 4px;
}

.prog-seats {
    color: var(--text-muted);
    font-size: 0.85rem;
    font-weight: 500;
}

@media (max-width: 768px) {
    .program-row {
        grid-template-columns: 1fr;
        gap: 16px;
        padding: 24px;
    }
    .prog-meta { text-align: left; }
    .prog-number { display: none; }
}

/* FACULTY */
.faculty-list {
    display: flex;
    flex-direction: column;
    gap: 16px;
    max-width: 900px;
    margin: 0 auto;
}
```

```
.faculty-row {
  display: flex;
  align-items: center;
  gap: 24px;
  padding: 24px 32px;
  background: var(--bg-card);
  border: 1px solid var(--border);
  border-radius: 16px;
  transition: var(--transition);
  cursor: pointer;
}

.faculty-row:hover {
  border-color: var(--primary);
  transform: translateX(4px);
  box-shadow: 0 12px 30px rgba(0,0,0,0.04);
}

.faculty-avatar {
  width: 56px; height: 56px; border-radius: 50%;
  display: grid; place-items: center;
  font-size: 1.1rem; font-weight: 700; font-family: 'Outfit',
sans-serif;
  flex-shrink: 0;
  color: #fff;
}

.faculty-info {
  flex: 1;
  display: flex;
  flex-direction: column;
  gap: 4px;
}

.faculty-row h4 { font-family: 'Outfit', sans-serif; font-weight: 700;
font-size: 1.15rem; color: var(--text-main); margin: 0; }

.faculty-meta {
  display: flex;
```

```

        justify-content: space-between;
        align-items: center;
        width: 100%;
    }

    .designation { color: var(--primary); font-size: 0.9rem; font-weight:
600; margin: 0; }
    .specialization { color: var(--text-muted); font-size: 0.85rem;
font-weight: 500; }

@media (max-width: 600px) {
    .faculty-row {
        flex-direction: column;
        align-items: flex-start;
        gap: 16px;
    }
    .faculty-meta {
        flex-direction: column;
        align-items: flex-start;
        gap: 8px;
    }
}

/* STUDENT BANNER */
.student-banner {
    position: relative; z-index: 1;
    margin: 0 5% 80px;
    background: linear-gradient(135deg, rgba(59,130,246,0.18),
rgba(56,189,248,0.1));
    border: 1px solid rgba(59,130,246,0.35);
    border-radius: 24px; padding: 56px 60px;
    display: flex; align-items: center; justify-content: space-between;
    gap: 40px; overflow: hidden;
}

.student-banner::before {
    content: '🎓'; position: absolute; right: 60px;
    top: 50%; transform: translateY(-50%);
    font-size: 140px; opacity: 0.07; pointer-events: none;
}

```

```
.banner-content h2 {
  font-family: 'Outfit', sans-serif;
  font-size: clamp(1.4rem, 3vw, 2rem); font-weight: 800;
margin-bottom: 12px;
}

.banner-content p { color: var(--text-muted); font-size: 0.95rem;
line-height: 1.7; max-width: 500px; }

/* FOOTER */
footer {
  position: relative; z-index: 1;
  background: #f8fafc;
  border-top: 1px solid var(--border);
  padding: 56px 5% 28px;
}

.footer-grid {
  display: grid; grid-template-columns: 2fr 1fr 1fr 1fr;
  gap: 48px; margin-bottom: 48px;
}

.footer-brand p { color: var(--text-muted); font-size: 0.88rem;
line-height: 1.7; margin-top: 14px; max-width: 280px; }

.footer-col h5 { font-family: 'Outfit', sans-serif; font-weight: 700;
font-size: 0.95rem; margin-bottom: 16px; }
.footer-col ul { list-style: none; display: flex; flex-direction:
column; gap: 10px; }
.footer-col ul a { color: var(--text-muted); text-decoration: none;
font-size: 0.88rem; transition: var(--transition); }
.footer-col ul a:hover { color: var(--primary); }

.footer-bottom {
  border-top: 1px solid var(--border); padding-top: 24px;
  display: flex; align-items: center; justify-content: space-between;
  color: var(--text-muted); font-size: 0.82rem;
}
```



```

/* REVEAL */
.reveal { opacity: 0; transform: translateY(40px); transition: opacity
0.7s ease, transform 0.7s ease; }
.reveal.visible { opacity: 1; transform: translateY(0); }

@media (max-width: 900px) {
  #about { grid-template-columns: 1fr; }
  .footer-grid { grid-template-columns: 1fr 1fr; }
  .student-banner { flex-direction: column; text-align: center; }
}

@media (max-width: 600px) {
  .nav-links { display: none; }
  .footer-grid { grid-template-columns: 1fr; }
  .hero-btns { flex-direction: column; align-items: center; }
}
</style>
<script src="https://unpkg.com/lucide@latest"></script>
<style>
  i[data-lucide] {
    width: 1.2em;
    height: 1.2em;
    vertical-align: text-bottom;
    display: inline-block;
  }
</style>
</head>
<body>

<!-- NAVBAR -->
<nav id="main-nav">
  <a href="index.html" class="nav-logo">
    <div class="logo-icon">S</div>
    <span>SIT Nagpur</span>
  </a>
  <ul class="nav-links">
    <li><a href="#about" class="active">About</a></li>
    <li><a href="#programs">Programs</a></li>
    <li><a href="#faculty">Faculty</a></li>
    <li><a href="students.html" class="nav-cta">Student Info</a></li>
  </ul>

```

```

    </ul>
</nav>

<!-- HERO -->
<section class="hero">
    <div class="hero-badge">Affiliated to RTMNU &middot; AICTE
Approved</div>
    <h1>
        Department of<br>
        <span class="highlight">Computer Science</span><br>
        & Engineering
    </h1>
    <p>
        Empowering the next generation of innovators at
        <strong>Symbiosis Institute of Technology, Nagpur</strong>.
        Excellence in education, research, and industry collaboration.
    </p>
    <div class="hero-btns">
        <a href="students.html" class="btn-primary" id="student-info-btn"><i
data-lucide="graduation-cap"></i> Student Information</a>
        <a href="#programs" class="btn-secondary"><i
data-lucide="book-open"></i> Explore Programs</a>
    </div>
</section>

<!-- STATS -->
<div class="stats-strip">
    <div class="stat-item">
        <span class="stat-num">1200+</span>
        <span class="stat-label">Students Enrolled</span>
    </div>
    <div class="stat-item">
        <span class="stat-num">48</span>
        <span class="stat-label">Expert Faculty</span>
    </div>
    <div class="stat-item">
        <span class="stat-num">95%</span>
        <span class="stat-label">Placement Rate</span>
    </div>
    <div class="stat-item">

```

```

        <span class="stat-num">25+</span>
        <span class="stat-label">Years of Excellence</span>
    </div>
    <div class="stat-item">
        <span class="stat-num">150+</span>
        <span class="stat-label">Research Papers</span>
    </div>
</div>

<!-- ABOUT -->
<section id="about">
    <div class="about-visual reveal">
        <div class="about-card-main">
            <div class="about-icon"><i data-lucide="lightbulb"></i></div>
            <h3>Vision of the Department</h3>
            <p>
                To be a globally recognized centre of excellence in Computer
Science
                & Engineering education, producing competent graduates who
contribute
                meaningfully to society through innovation and ethical practice.
            </p>
        </div>
        <div class="floating-badge top-right"><i data-lucide="trophy"></i>
NAAC Accredited</div>
        <div class="floating-badge bot-left"><i data-lucide="star"></i> NBA
Approved</div>
    </div>

    <div class="about-text reveal">
        <div class="section-tag">About Us</div>
        <h2 class="section-title">Shaping Tomorrow's<br>Tech Leaders</h2>
        <p class="section-sub" style="max-width:100%; margin-bottom:28px;">
            The Department of Computer Science & Engineering at SIT Nagpur
has been
            at the forefront of technological education since its
establishment. We foster
            a culture of learning, research, and innovation.
        </p>
        <ul class="feature-list">

```

```

        <li>State-of-the-art Computing Laboratories</li>
        <li>Industry-integrated Curriculum & Live Projects</li>
        <li>Research Collaborations with IITs & NITs</li>
        <li>Active Placement Cell with 200+ Company Tie-ups</li>
        <li>Student Clubs: Coding, AI/ML, Robotics, Cybersecurity</li>
        <li>Merit Scholarships & Financial Aid Programs</li>
    </ul>
</div>
</section>

<!-- PROGRAMS -->
<section id="programs">
    <div style="text-align:center; margin-bottom:56px;">
        <div class="section-tag">Academics</div>
        <h2 class="section-title">Our Programs</h2>
        <p class="section-sub" style="margin:0 auto;">
            Choose from a diverse range of undergraduate and postgraduate
programs
            designed to meet global industry standards.
        </p>
    </div>
    <div class="programs-list">
        <div class="program-row reveal">
            <div class="prog-number">01</div>
            <div class="prog-content">
                <h3>B.E. Computer Science & Engineering</h3>
                <p>Comprehensive 4-year program covering algorithms, OS,
networks, databases, AI/ML, and software engineering.</p>
            </div>
            <div class="prog-meta">
                <span class="prog-duration">4 Years</span>
                <span class="prog-seats">60 Seats</span>
            </div>
        </div>
        <div class="program-row reveal">
            <div class="prog-number">02</div>
            <div class="prog-content">
                <h3>B.E. AI & Machine Learning</h3>
                <p>Specialized program focusing on deep learning, data science,
computer vision, and intelligent systems.</p>
            </div>
        </div>
    </div>

```

```
</div>
<div class="prog-meta">
  <span class="prog-duration">4 Years</span>
  <span class="prog-seats">60 Seats</span>
</div>
</div>
<div class="program-row reveal">
  <div class="prog-number">03</div>
  <div class="prog-content">
    <h3>B.E. Cybersecurity</h3>
    <p>Modern curriculum covering ethical hacking, network security,
cryptography, and digital forensics.</p>
  </div>
  <div class="prog-meta">
    <span class="prog-duration">4 Years</span>
    <span class="prog-seats">30 Seats</span>
  </div>
</div>
<div class="program-row reveal">
  <div class="prog-number">04</div>
  <div class="prog-content">
    <h3>B.E. Information Technology</h3>
    <p>Broad-based program in software development, web
technologies, cloud computing, and enterprise systems.</p>
  </div>
  <div class="prog-meta">
    <span class="prog-duration">4 Years</span>
    <span class="prog-seats">60 Seats</span>
  </div>
</div>
<div class="program-row reveal">
  <div class="prog-number">05</div>
  <div class="prog-content">
    <h3>M.Tech Computer Science</h3>
    <p>Advanced 2-year postgraduate program with research
opportunities and specializations in cutting-edge areas.</p>
  </div>
  <div class="prog-meta">
    <span class="prog-duration">2 Years</span>
    <span class="prog-seats">24 Seats</span>
```

```

    </div>
  </div>
  <div class="program-row reveal">
    <div class="prog-number">06</div>
    <div class="prog-content">
      <h3>M.Tech Data Science</h3>
      <p>Data-driven postgraduate program with hands-on industry
projects, big data, and analytics specialization.</p>
    </div>
    <div class="prog-meta">
      <span class="prog-duration">2 Years</span>
      <span class="prog-seats">24 Seats</span>
    </div>
  </div>
</div>
</section>

<!-- FACULTY -->
<section id="faculty">
  <div style="text-align:center; margin-bottom:56px;">
    <div class="section-tag">Our Team</div>
    <h2 class="section-title">Meet the Faculty</h2>
    <p class="section-sub" style="margin:0 auto;">
      Learn from experienced academicians and industry experts who are
passionate
      about shaping your future.
    </p>
  </div>
  <div class="faculty-list">
    <div class="faculty-row reveal">
      <div class="faculty-avatar"
style="background:linear-gradient(135deg,#3b82f6,#2563eb);
color:#fff;">DR</div>
      <div class="faculty-info">
        <h4>Dr. Rajesh Sharma</h4>
        <div class="faculty-meta">
          <span class="designation">Head of Department</span>
          <span class="specialization">Ph.D & middot; AI & amp; Machine
Learning</span>
        </div>
      </div>
    </div>
  </div>

```

```

        </div>
    </div>
    <div class="faculty-row reveal">
        <div class="faculty-avatar"
style="background:linear-gradient(135deg,#38bdf8,#0284c7);
color:#fff;">PM</div>
        <div class="faculty-info">
            <h4>Prof. Priya Mehta</h4>
            <div class="faculty-meta">
                <span class="designation">Associate Professor</span>
                <span class="specialization">M.Tech &middot; Data
Science</span>
            </div>
        </div>
    </div>
    <div class="faculty-row reveal">
        <div class="faculty-avatar"
style="background:linear-gradient(135deg,#f43f5e,#e11d48);
color:#fff;">AK</div>
        <div class="faculty-info">
            <h4>Prof. Amit Kumar</h4>
            <div class="faculty-meta">
                <span class="designation">Assistant Professor</span>
                <span class="specialization">Ph.D &middot;
Cybersecurity</span>
            </div>
        </div>
    </div>
    <div class="faculty-row reveal">
        <div class="faculty-avatar"
style="background:linear-gradient(135deg,#fbbf24,#f59e0b);
color:#1a1400;">SP</div>
        <div class="faculty-info">
            <h4>Dr. Sneha Patil</h4>
            <div class="faculty-meta">
                <span class="designation">Associate Professor</span>
                <span class="specialization">Ph.D &middot; Cloud
Computing</span>
            </div>
        </div>
    </div>

```

```

        </div>
    </div>
</section>

<!-- STUDENT BANNER -->
<div class="student-banner reveal">
    <div class="banner-content">
        <h2><i data-lucide="graduation-cap"></i> Access Student Information
Portal</h2>
        <p>
            View student profiles, academic records, attendance, results, and
            important notices – all in one place. Click below to explore the
            student information dashboard.
        </p>
    </div>
    <a href="students.html" class="btn-primary" id="goto-students-btn"
style="flex-shrink:0;">
        View Students <i data-lucide="arrow-right"></i>
    </a>
</div>

<!-- FOOTER -->
<footer>
    <div class="footer-grid">
        <div class="footer-brand">
            <a href="index.html" class="nav-logo" style="display:inline-flex;
margin-bottom:8px;">
                <div class="logo-icon">S</div>
                <span>SIT Nagpur</span>
            </a>
            <p>Symbiosis Institute of Technology, Nagpur.<br>
                Department of Computer Science & Engineering.<br>
                Committed to excellence in technical education.</p>
        </div>
        <div class="footer-col">
            <h5>Quick Links</h5>
            <ul>
                <li><a href="#about">About Us</a></li>
                <li><a href="#programs">Programs</a></li>
                <li><a href="#faculty">Faculty</a></li>

```



```

        <li><a href="students.html">Student Info</a></li>
    </ul>
</div>
<div class="footer-col">
    <h5>Resources</h5>
    <ul>
        <li><a href="#">Library</a></li>
        <li><a href="#">E-Learning</a></li>
        <li><a href="#">Research Lab</a></li>
        <li><a href="#">Placement Cell</a></li>
    </ul>
</div>
<div class="footer-col">
    <h5>Contact</h5>
    <ul>
        <li><a href="#"><i data-lucide="map-pin"></i> Nagpur,
Maharashtra</a></li>
        <li><a href="#"><i data-lucide="phone"></i> +91 712 000
0000</a></li>
        <li><a href="#"><i data-lucide="mail"></i>
cse@sitnagpur.edu.in</a></li>
    </ul>
</div>
</div>
<div class="footer-bottom">
    <span>&copy; 2026 SIT Nagpur &mdash; Department of CSE. All rights
reserved.</span>
    <span>Designed with <i data-lucide="heart"></i> for
excellence</span>
</div>
</footer>

<script>
    // =====
    // JAVASCRIPT - Welcome Alert
    // =====
    window.addEventListener('load', function () {
        alert('Welcome to SIT Nagpur');
    });

```

```

// =====
// Scroll Reveal Animation
// =====
const revealObserver = new IntersectionObserver(
  (entries) => {
    entries.forEach((entry) => {
      if (entry.isIntersecting) {
        entry.target.classList.add('visible');
        revealObserver.unobserve(entry.target);
      }
    });
  },
  { threshold: 0.12 }
);

document.querySelectorAll('.reveal').forEach((el) =>
revealObserver.observe(el));

// =====
// Active Nav on Scroll
// =====
const navLinks = document.querySelectorAll('.nav-links a');
const navObserver = new IntersectionObserver(
  (entries) => {
    entries.forEach((entry) => {
      if (entry.isIntersecting) {
        navLinks.forEach((link) => {
          link.classList.remove('active');
          if (link.getAttribute('href') === '#' + entry.target.id) {
            link.classList.add('active');
          }
        });
      }
    });
  },
  { threshold: 0.4 }
);

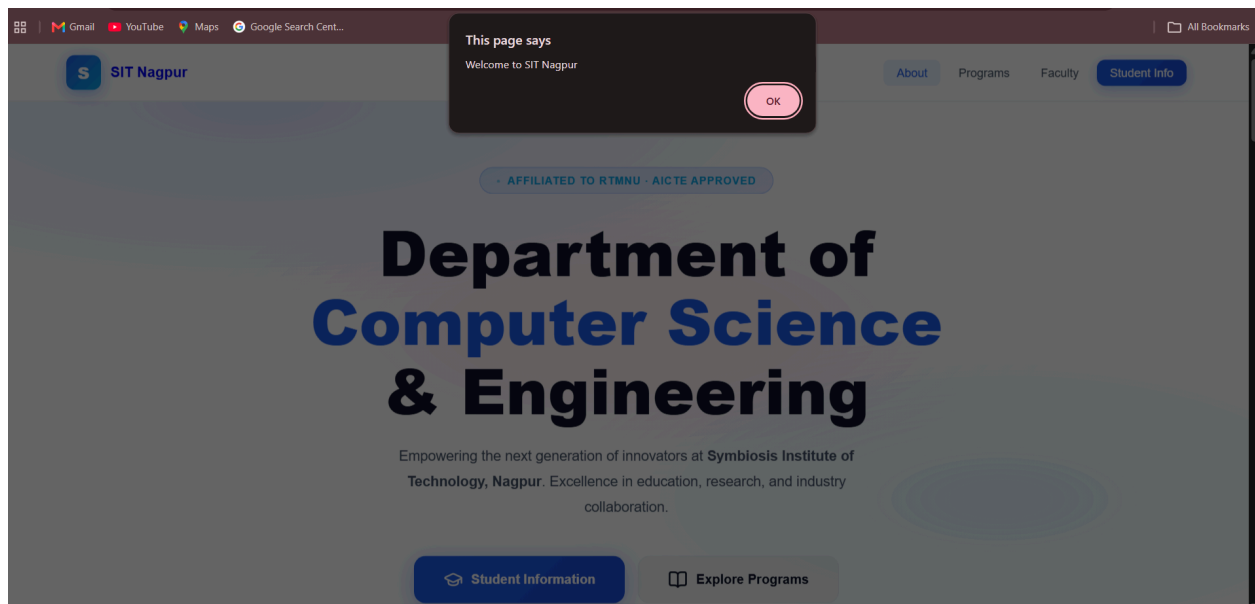
document.querySelectorAll('section[id]').forEach((sec) =>
navObserver.observe(sec));

// =====

```

```
// Smooth Scroll for Anchor Links
// =====
document.querySelectorAll('a[href^="#"]').forEach((anchor) => {
  anchor.addEventListener('click', function (e) {
    const target = document.querySelector(this.getAttribute('href'));
    if (target) {
      e.preventDefault();
      target.scrollIntoView({ behavior: 'smooth', block: 'start' });
    }
  });
});
</script>
<script>
  lucide.createIcons();
</script>
</body>
</html>
```

Output:



Result/Conclusion:

The case study was successfully implemented using **HTML** and **JavaScript**. Different methods of including JavaScript—**Inline**, **Internal**, and **External**—were demonstrated effectively. Alert boxes were used to display welcome messages, while console methods were used for debugging and monitoring program execution. The application successfully displayed student information and responded to user interactions, providing a clear understanding of the basic concepts of client-side web development.

Case Study No.: 2

Case Study Title

Online Shopping Billing System using HTML, CSS, HTML5, and JavaScript

Theory

An **Online Shopping Billing System** is a web-based application that automates the process of generating customer bills for purchased products. It collects customer details, product information, pricing, discounts, taxes, and payment details, then calculates the final payable amount accurately. Such systems reduce manual calculation errors, save time, and improve billing efficiency in retail stores, supermarkets, and e-commerce platforms.

This case study is developed using **HTML5**, **CSS3**, and **JavaScript**.

HTML5

HTML5 is used to create the structure of the webpage. It provides form elements such as text boxes, number fields, and drop-down lists to collect customer information including customer name, mobile number, invoice number, product name, quantity, rate per kilogram, discount, GST, packing charges, payment mode, and membership status.

CSS3

CSS3 is used to design the user interface of the application. It improves the appearance of the webpage by applying colors, fonts, spacing, borders, buttons, and responsive layouts, making the billing system more attractive and user-friendly.

JavaScript

JavaScript is used to add functionality to the webpage. It accepts user inputs, performs calculations, applies business logic, and dynamically generates the shopping invoice without refreshing the page.

Billing Calculation

The billing system performs the following calculations:

- **Total Amount**

$$\text{Total} = \text{Quantity} \times \text{Rate per Kg}$$

- **Discount Amount**

$$\text{Discount Amount} = (\text{Total} \times \text{Discount \%}) \div 100$$

- **Amount after Discount**

$$\text{Amount after Discount} = \text{Total} - \text{Discount Amount}$$

- **GST Amount**

$$\text{GST Amount} = (\text{Amount after Discount} \times \text{GST \%}) \div 100$$

- **Final Bill**

$$\text{Final Bill} = \text{Amount after Discount} + \text{GST Amount} + \text{Packing Charges}$$

- If the customer has a **Membership**, an additional discount (for example ₹100) is deducted from the final bill.

Features of the System

- Accepts customer and product details.
- Calculates the total purchase amount.
- Applies discount and GST automatically.
- Adds packing charges to the bill.
- Supports multiple payment methods such as **Cash**, **Card**, and **UPI**.
- Provides membership benefits by applying an additional discount.
- Generates a detailed invoice dynamically on the webpage.
- Simple, accurate, and easy to use.

Applications

- Grocery Stores
- Supermarkets
- Retail Shops
- E-commerce Websites
- Departmental Stores

- Wholesale Shops

Advantages

- Reduces manual calculation errors.
- Generates bills quickly and accurately.
- Improves customer service.
- Saves time and effort.
- Provides a professional billing system.
- Easy to modify and maintain.
- Enhances user experience through an interactive web interface.

1. Case Study Program Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width,initial-scale=1">

<title>Online Shopping Billing System</title>

<style>

body{
font-family:Arial;
background:#eef5ff;
padding:40px;
}

.container{
max-width:700px;
margin:auto;
background:white;
padding:25px;
border-radius:10px;
box-shadow:0 0 15px gray;
```



```
}

h2{
text-align:center;
color:#0b5394;
}

label{
font-weight:bold;
}

input,select{
width:100%;
padding:10px;
margin-top:5px;
margin-bottom:15px;
}

button{
width:100%;
padding:12px;
background:#0b5394;
color:white;
border:none;
font-size:18px;
cursor:pointer;
}

button:hover{
background:#073763;
}

#bill{
margin-top:25px;
background:#f4f4f4;
padding:20px;
border-radius:8px;
}

</style>
```

```
</head>

<body>

<div class="container">

<h2>Online Shopping Billing System</h2>

<label>Customer Name</label>
<input type="text" id="name">

<label>Mobile Number</label>
<input type="tel" id="mobile">

<label>Invoice Number</label>
<input type="text" id="invoice">

<label>Product Name</label>
<input type="text" id="product">

<label>Quantity (Kg)</label>
<input type="number" id="qty">

<label>Rate per Kg (₹)</label>
<input type="number" id="rate">

<label>Discount (%)</label>
<input type="number" id="discount">

<label>GST (%)</label>
<input type="number" id="gst">

<label>Packing Charges (₹)</label>
<input type="number" id="packing">

<label>Payment Mode</label>

<select id="payment">
<option>Cash</option>
```

```
<option>Card</option>
<option>UPI</option>
</select>

<label>Membership</label>

<select id="member">
<option>No</option>
<option>Yes</option>
</select>

<button onclick="calculateBill()">Generate Bill</button>

<div id="bill"></div>

</div>

<script>

function calculateBill(){

let name=document.getElementById("name").value;
let mobile=document.getElementById("mobile").value;
let invoice=document.getElementById("invoice").value;
let product=document.getElementById("product").value;

let qty=Number(document.getElementById("qty").value);
let rate=Number(document.getElementById("rate").value);
let discount=Number(document.getElementById("discount").value);
let gst=Number(document.getElementById("gst").value);
let packing=Number(document.getElementById("packing").value);

let payment=document.getElementById("payment").value;
let member=document.getElementById("member").value;

let total=qty*rate;

let discountAmount=(total*discount)/100;

let subtotal=total-discountAmount;
```

```
let gstAmount=(subtotal*gst)/100;

let finalBill=subtotal+gstAmount+packing;

if (member=="Yes") {

finalBill=finalBill-100;

}

document.getElementById("bill").innerHTML=`

<h3>Invoice</h3>

<b>Customer :</b> ${name}<br>
<b>Mobile :</b> ${mobile}<br>
<b>Invoice :</b> ${invoice}<br>
<b>Product :</b> ${product}<br>
<b>Quantity :</b> ${qty} Kg<br>
<b>Rate :</b> ₹${rate}<br>
<b>Total :</b> ₹${total.toFixed(2)}<br>
<b>Discount :</b> ₹${discountAmount.toFixed(2)}<br>
<b>GST :</b> ₹${gstAmount.toFixed(2)}<br>
<b>Packing :</b> ₹${packing}<br>
<b>Payment :</b> ${payment}<br>
<b>Membership :</b> ${member}<br>

<hr>

<h2>Final Bill : ₹${finalBill.toFixed(2)}</h2>

`;

}

</script>

</body>
</html>
```

Output:

Online Shopping Billing System

Customer Name

RADHIKA SURESHRAO CHAUDHARI

Mobile Number

9405606890

Invoice Number

32

Product Name

Mac

Quantity (Kg)

1

Rate per Kg (₹)

1000000

Discount (%)

20

GST (%)

20

Packing Charges (₹)

5000

Payment Mode

UPI



Membership

No



Generate Bill



Result/Conclusion:

The experiment and case study were successfully implemented using **HTML5, CSS3, and JavaScript**. Primitive data types such as **String, Number, Boolean, Undefined, and Null** were demonstrated along with type conversion. An **Online Shopping Billing System** was developed that accepts customer and product details, calculates the total amount after applying discount, GST, packing charges, and membership benefits, and generates a detailed invoice. This case study improved the understanding of HTML form elements, JavaScript data types, arithmetic operations, user input handling, and dynamic webpage generation.

Case Study No.: 3

Case Study Title

Scholarship Eligibility Checker using HTML5, CSS3, and JavaScript

Theory

A **Scholarship Eligibility Checker** is a web-based application that helps students determine their scholarship category based on the marks they have obtained. The application accepts the student's marks as input, validates the entered value, and displays the appropriate scholarship category according to predefined eligibility criteria.

This case study is developed using **HTML5**, **CSS3**, and **JavaScript** to create an attractive, responsive, and interactive web application.

HTML5

HTML5 provides the structure of the webpage. It is used to create input fields, buttons, headings, labels, and containers that allow the user to enter marks and display the scholarship result. Modern HTML5 input types such as **number** are used to ensure valid numerical input.

CSS3

CSS3 is used to design the user interface of the application. It enhances the webpage by applying colors, gradients, animations,

shadows, icons, and responsive layouts. A modern dark theme with interactive cards improves the overall user experience and makes the application visually appealing.

JavaScript

JavaScript adds functionality to the webpage. It reads the user's marks, validates the input, checks the eligibility criteria using conditional statements (**if-else**), and dynamically displays the scholarship category along with an appropriate message without reloading the webpage.

Working of the Application

1. The user enters marks between **0 and 100**.
2. JavaScript validates the entered value.
3. The program compares the marks with the scholarship criteria.
4. The corresponding scholarship category is determined.
5. The result is displayed with suitable colors, icons, and messages.

Features

- Accepts marks using an HTML5 number input.
- Validates user input.
- Uses conditional statements (**if-else**) for decision making.
- Displays scholarship category dynamically.
- Responsive and attractive user interface.
- Modern dark theme with animations and icons.
- User-friendly and easy to operate.

Applications

- Schools
- Colleges
- Universities
- Scholarship Portals
- Educational Institutions
- Online Admission Systems

Advantages

- Provides instant scholarship eligibility results.
- Eliminates manual evaluation.
- Reduces calculation errors.
- Easy to use and maintain.
- Responsive interface for different devices.
- Improves user experience through dynamic content.

1. Case Study Program Code:

index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Scholarship Eligibility Checker -
Enter your marks to find your scholarship category">
  <title>Scholarship Eligibility Checker</title>
  <link rel="stylesheet" href="style.css">
  <link
href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;
800&display=swap" rel="stylesheet">
</head>
<body>
  <div class="background-orbs">
```

```

        <div class="orb orb-1"></div>
        <div class="orb orb-2"></div>
        <div class="orb orb-3"></div>
    </div>

    <div class="container">
        <header class="header">
            <div class="logo">
                <span class="logo-icon">
                    <!-- Graduation Cap -->
                    <svg viewBox="0 0 48 48" fill="none"
xmlns="http://www.w3.org/2000/svg">
                        <path d="M24 6L2 18L24 30L46 18L24 6Z"
fill="#e8e8ed" stroke="#ffffff" stroke-width="1"/>
                        <path d="M12 24V36L24 42L36 36V24L24 30L12 24Z"
fill="#b0b0be" stroke="#d0d0d8" stroke-width="0.5"/>
                        <line x1="40" y1="18" x2="40" y2="38"
stroke="#d0d0d8" stroke-width="1.5" stroke-linecap="round"/>
                        <path d="M37 38C37 36.3 38.3 35 40 35C41.7 35 43
36.3 43 38C43 39.7 41.7 42 40 42C38.3 42 37 39.7 37 38Z" fill="#d0d0d8"/>
                    </svg>
                </span>
                <h1>Scholarship Eligibility</h1>
            </div>
            <p class="subtitle">Enter your marks to discover your
scholarship category</p>
        </header>

        <main class="main-card" id="mainCard">
            <div class="input-section">
                <label for="marksInput" class="input-label">Enter Your
Marks</label>
                <div class="input-wrapper">
                    <input
                        type="number"
                        id="marksInput"
                        min="0"
                        max="100"
                        placeholder="0 - 100"
                        autocomplete="off"

```

```

        >
        <span class="input-suffix">/ 100</span>
    </div>
    <button id="checkBtn" onclick="checkScholarship()">
        <span>Check Eligibility</span>
        <svg width="20" height="20" viewBox="0 0 24 24"
fill="none" stroke="currentColor" stroke-width="2.5"
stroke-linecap="round" stroke-linejoin="round">
            <path d="M5 12h14M12 5l7 7 7"/>
        </svg>
    </button>
</div>

<div class="result-section" id="resultSection">
    <div class="result-card" id="resultCard">
        <div class="result-icon" id="resultIcon"></div>
        <h2 class="result-title" id="resultTitle"></h2>
        <p class="result-marks" id="resultMarks"></p>
        <p class="result-message" id="resultMessage"></p>
    </div>
</div>
</main>

<section class="criteria-section">
    <h2 class="criteria-heading">Scholarship Criteria</h2>
    <div class="criteria-grid">
        <div class="criteria-item not-eligible">
            <div class="criteria-bar"></div>
            <div class="criteria-icon">
                <!-- Shield with X -->
                <svg viewBox="0 0 24 24" fill="none"
xmlns="http://www.w3.org/2000/svg">
                    <path d="M12 2L4 6V12C4 16.42 7.4 20.56 12
22C16.6 20.56 20 16.42 20 12V6L12 2Z" fill="#2a1215" stroke="#ef4444"
stroke-width="1.5"/>
                    <path d="M9.5 9.5L14.5 14.5M14.5 9.5L9.5 14.5"
stroke="#ef4444" stroke-width="2" stroke-linecap="round"/>
                </svg>
            </div>
            <div class="criteria-range">Below 50</div>
        </div>
    </div>
</section>

```

```

        <div class="criteria-label">Not Eligible</div>
    </div>
    <div class="criteria-item bronze">
        <div class="criteria-bar"></div>
        <div class="criteria-icon">
            <!-- Bronze Medal -->
            <svg viewBox="0 0 24 24" fill="none"
xmlns="http://www.w3.org/2000/svg">
                <path d="M8 2H16L14.5 7H9.5L8 2Z"
fill="#a0522d"/>
                <path d="M10 2L8 7" stroke="#8b4513"
stroke-width="0.5"/>
                <path d="M14 2L16 7" stroke="#8b4513"
stroke-width="0.5"/>
                <circle cx="12" cy="14.5" r="7" fill="#cd7f32"
stroke="#a0622a" stroke-width="0.8"/>
                <circle cx="12" cy="14.5" r="5.2" fill="none"
stroke="#daa06d" stroke-width="0.6"/>
                <path d="M10 13L11.5 15L14.5 11.5"
stroke="#fff" stroke-width="1.5" stroke-linecap="round"
stroke-linejoin="round"/>
            </svg>
        </div>
        <div class="criteria-range">50 - 69</div>
        <div class="criteria-label">Bronze</div>
    </div>
    <div class="criteria-item silver">
        <div class="criteria-bar"></div>
        <div class="criteria-icon">
            <!-- Silver Medal -->
            <svg viewBox="0 0 24 24" fill="none"
xmlns="http://www.w3.org/2000/svg">
                <path d="M8 2H16L14.5 7H9.5L8 2Z"
fill="#8a8a98"/>
                <path d="M10 2L8 7" stroke="#707080"
stroke-width="0.5"/>
                <path d="M14 2L16 7" stroke="#707080"
stroke-width="0.5"/>
                <circle cx="12" cy="14.5" r="7" fill="#c0c0c0"
stroke="#a8a8b0" stroke-width="0.8"/>

```

```

        <circle cx="12" cy="14.5" r="5.2" fill="none"
stroke="#d8d8e0" stroke-width="0.6"/>
        <path d="M10 13L11.5 15L14.5 11.5"
stroke="#fff" stroke-width="1.5" stroke-linecap="round"
stroke-linejoin="round"/>
    </svg>
</div>
<div class="criteria-range">70 - 84</div>
<div class="criteria-label">Silver</div>
</div>
<div class="criteria-item gold">
    <div class="criteria-bar"></div>
    <div class="criteria-icon">
        <!-- Gold Medal -->
        <svg viewBox="0 0 24 24" fill="none"
xmlns="http://www.w3.org/2000/svg">
            <path d="M8 2H16L14.5 7H9.5L8 2Z"
fill="#daa520"/>
            <path d="M10 2L8 7" stroke="#b8860b"
stroke-width="0.5"/>
            <path d="M14 2L16 7" stroke="#b8860b"
stroke-width="0.5"/>
            <circle cx="12" cy="14.5" r="7" fill="#ffd700"
stroke="#daa520" stroke-width="0.8"/>
            <circle cx="12" cy="14.5" r="5.2" fill="none"
stroke="#ffe44d" stroke-width="0.6"/>
            <path d="M10 13L11.5 15L14.5 11.5"
stroke="#fff" stroke-width="1.5" stroke-linecap="round"
stroke-linejoin="round"/>
        </svg>
    </div>
<div class="criteria-range">85 - 94</div>
<div class="criteria-label">Gold</div>
</div>
<div class="criteria-item platinum">
    <div class="criteria-bar"></div>
    <div class="criteria-icon">
        <!-- Diamond/Trophy -->
        <svg viewBox="0 0 24 24" fill="none"
xmlns="http://www.w3.org/2000/svg">

```

```

        <path d="M5 4H19L21 8L12 20L3 8L5 4Z"
fill="#e5e4e2" stroke="#c8c8c6" stroke-width="0.8"/>
        <path d="M3 8H21" stroke="#d0d0ce"
stroke-width="0.6"/>
        <path d="M12 20L8.5 8L10 4" stroke="#c8c8c6"
stroke-width="0.5"/>
        <path d="M12 20L15.5 8L14 4" stroke="#c8c8c6"
stroke-width="0.5"/>
        <path d="M8.5 8L10 4H14L15.5 8" fill="#f0efed"
stroke="none"/>
    </svg>
</div>
<div class="criteria-range">95 - 100</div>
<div class="criteria-label">Platinum</div>
</div>
</div>
</section>
</div>

<script src="script.js"></script>
</body>
</html>

```

script.js:

```

// Scholarship Eligibility Checker
// Case Study: A college awards scholarships based on marks

// Detailed custom SVG icons for each scholarship tier
const ICONS = {
  'not-eligible': `
    <svg viewBox="0 0 64 64" fill="none"
xmlns="http://www.w3.org/2000/svg">
      <!-- Shield body -->
      <path d="M32 4L8 16V32C8 44.84 18.8 56.12 32 60C45.2 56.12 56
44.84 56 32V16L32 4Z" fill="#2a1215" stroke="#ef4444" stroke-width="2.5"/>
      <!-- Inner shield highlight -->
      <path d="M32 10L14 19V32C14 42.2 22.4 51.2 32 54.4C41.6 51.2
50 42.2 50 32V19L32 10Z" fill="#1f0a0d" stroke="#ef4444"
stroke-width="0.5" opacity="0.4"/>

```

```

        <!-- X mark -->
        <path d="M23 23L41 41" stroke="#ef4444" stroke-width="4.5"
stroke-linecap="round"/>
        <path d="M41 23L23 41" stroke="#ef4444" stroke-width="4.5"
stroke-linecap="round"/>
        <!-- Corner accents -->
        <circle cx="32" cy="14" r="1.5" fill="#ef4444" opacity="0.5"/>
</svg>`,

```

```

'bronze': `
    <svg viewBox="0 0 64 64" fill="none"
xmlns="http://www.w3.org/2000/svg">
        <!-- Ribbon -->
        <path d="M22 4H42L38 18H26L22 4Z" fill="#a0522d"/>
        <path d="M26 4L22 18" stroke="#8b4513" stroke-width="0.8"/>
        <path d="M32 4V18" stroke="#8b4513" stroke-width="0.5"
opacity="0.4"/>
        <path d="M38 4L42 18" stroke="#8b4513" stroke-width="0.8"/>
        <!-- Medal body -->
        <circle cx="32" cy="38" r="20" fill="#cd7f32" stroke="#a0622a"
stroke-width="2"/>
        <!-- Inner ring -->
        <circle cx="32" cy="38" r="15" fill="none" stroke="#daa06d"
stroke-width="1.2"/>
        <!-- Star in center -->
        <polygon points="32,25 35,33 43,33 37,38 39,46 32,42 25,46
27,38 21,33 29,33" fill="#e8a848" stroke="#b8762a" stroke-width="0.5"/>
        <!-- Shine -->
        <path d="M22 28C24 24 28 22 32 22" stroke="#e8c088"
stroke-width="1" stroke-linecap="round" opacity="0.6"/>
    </svg>`,

```

```

'silver': `
    <svg viewBox="0 0 64 64" fill="none"
xmlns="http://www.w3.org/2000/svg">
        <!-- Ribbon -->
        <path d="M22 4H42L38 18H26L22 4Z" fill="#8a8a98"/>
        <path d="M26 4L22 18" stroke="#707080" stroke-width="0.8"/>
        <path d="M32 4V18" stroke="#707080" stroke-width="0.5"
opacity="0.4"/>

```

```

        <path d="M38 4L42 18" stroke="#707080" stroke-width="0.8"/>
        <!-- Medal body -->
        <circle cx="32" cy="38" r="20" fill="#c0c0c0" stroke="#a8a8b0"
stroke-width="2"/>
        <!-- Inner ring -->
        <circle cx="32" cy="38" r="15" fill="none" stroke="#d8d8e0"
stroke-width="1.2"/>
        <!-- Star in center -->
        <polygon points="32,25 35,33 43,33 37,38 39,46 32,42 25,46
27,38 21,33 29,33" fill="#d8d8e0" stroke="#a0a0a8" stroke-width="0.5"/>
        <!-- Shine -->
        <path d="M22 28C24 24 28 22 32 22" stroke="#e8e8f0"
stroke-width="1" stroke-linecap="round" opacity="0.7"/>
    </svg>`,

    'gold': `
        <svg viewBox="0 0 64 64" fill="none"
xmlns="http://www.w3.org/2000/svg">
        <!-- Ribbon -->
        <path d="M22 4H42L38 18H26L22 4Z" fill="#daa520"/>
        <path d="M26 4L22 18" stroke="#b8860b" stroke-width="0.8"/>
        <path d="M32 4V18" stroke="#b8860b" stroke-width="0.5"
opacity="0.4"/>
        <path d="M38 4L42 18" stroke="#b8860b" stroke-width="0.8"/>
        <!-- Medal body -->
        <circle cx="32" cy="38" r="20" fill="#ffd700" stroke="#daa520"
stroke-width="2"/>
        <!-- Inner ring -->
        <circle cx="32" cy="38" r="15" fill="none" stroke="#ffe44d"
stroke-width="1.2"/>
        <!-- Star in center -->
        <polygon points="32,25 35,33 43,33 37,38 39,46 32,42 25,46
27,38 21,33 29,33" fill="#ffe866" stroke="#d4a800" stroke-width="0.5"/>
        <!-- Shine -->
        <path d="M22 28C24 24 28 22 32 22" stroke="#fff4a0"
stroke-width="1" stroke-linecap="round" opacity="0.8"/>
    </svg>`,

    'platinum': `

```



```

    <svg viewBox="0 0 64 64" fill="none"
xmlns="http://www.w3.org/2000/svg">
    <!-- Diamond top facets -->
    <path d="M12 20H52L32 56L12 20Z" fill="#e5e4e2"
stroke="#c8c8c6" stroke-width="1.5"/>
    <!-- Top edge -->
    <path d="M8 20L16 8H48L56 20H8Z" fill="#f0efed"
stroke="#d0d0ce" stroke-width="1"/>
    <!-- Center facets -->
    <path d="M8 20L32 56L24 20Z" fill="#d8d7d5"/>
    <path d="M56 20L32 56L40 20Z" fill="#d8d7d5"/>
    <!-- Top inner facets -->
    <path d="M16 8L24 20H8Z" fill="#e0dfdd"/>
    <path d="M48 8L40 20H56Z" fill="#e0dfdd"/>
    <path d="M16 8H48L40 20H24Z" fill="#f5f4f2"/>
    <!-- Center vertical line -->
    <path d="M32 56L32 20" stroke="#c8c8c6" stroke-width="0.6"/>
    <path d="M32 20L16 8" stroke="#c8c8c6" stroke-width="0.5"/>
    <path d="M32 20L48 8" stroke="#c8c8c6" stroke-width="0.5"/>
    <!-- Sparkle highlights -->
    <circle cx="28" cy="16" r="1" fill="#fff" opacity="0.8"/>
    <circle cx="38" cy="14" r="0.7" fill="#fff" opacity="0.6"/>
    <path d="M20 26L22 24" stroke="#fff" stroke-width="0.8"
stroke-linecap="round" opacity="0.5"/>
    </svg>`
};

```

```

function checkScholarship() {
    const input = document.getElementById('marksInput');
    const resultSection = document.getElementById('resultSection');
    const resultCard = document.getElementById('resultCard');
    const resultIcon = document.getElementById('resultIcon');
    const resultTitle = document.getElementById('resultTitle');
    const resultMarks = document.getElementById('resultMarks');
    const resultMessage = document.getElementById('resultMessage');

    const marks = parseFloat(input.value);

    // Validate input
    if (isNaN(marks) || marks < 0 || marks > 100) {

```

```

        input.classList.add('shake');
        setTimeout(() => input.classList.remove('shake'), 400);
        resultSection.classList.remove('visible');
        alert('Please enter valid marks between 0 and 100.');
```

return;

}

```

let category, message, cssClass;

if (marks < 50) {
    category = 'Not Eligible';
    message = 'Score below 50. Keep working hard – you can improve!';
    cssClass = 'not-eligible';
} else if (marks >= 50 && marks <= 69) {
    category = 'Bronze Scholarship';
    message = 'Great start! You\'ve earned a Bronze Scholarship.';
    cssClass = 'bronze';
} else if (marks >= 70 && marks <= 84) {
    category = 'Silver Scholarship';
    message = 'Impressive performance! You\'ve earned a Silver
Scholarship.';
    cssClass = 'silver';
} else if (marks >= 85 && marks <= 94) {
    category = 'Gold Scholarship';
    message = 'Outstanding! You\'ve earned a Gold Scholarship.';
    cssClass = 'gold';
} else if (marks >= 95 && marks <= 100) {
    category = 'Platinum Scholarship';
    message = 'Exceptional! You\'ve earned the prestigious Platinum
Scholarship!';
    cssClass = 'platinum';
}

// Update the result card
resultCard.className = 'result-card ' + cssClass;
resultIcon.innerHTML = ICONS[cssClass];
resultTitle.textContent = category;
resultMarks.textContent = 'Your Marks: ' + marks + ' / 100';
resultMessage.textContent = message;
```

```

    // Show result with animation
    resultSection.classList.remove('visible');
    // Force reflow to restart animation
    void resultSection.offsetWidth;
    resultSection.classList.add('visible');
}

// Allow pressing Enter to check
document.getElementById('marksInput').addEventListener('keydown', function
(e) {
    if (e.key === 'Enter') {
        checkScholarship();
    }
});

```

Style.css:

```

/* ===== Reset & Base ===== */
*,
*::before,
*::after {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

:root {
    --bg-primary: #0e0e12;
    --bg-card: rgba(255, 255, 255, 0.03);
    --bg-card-hover: rgba(255, 255, 255, 0.06);
    --text-primary: #e8e8ed;
    --text-secondary: #6e6e82;
    --border-color: rgba(255, 255, 255, 0.07);
    --white: #ffffff;
    --shadow-glow: 0 0 80px rgba(255, 255, 255, 0.03);

    /* Scholarship tier colors — used only on results */
    --not-eligible: #ef4444;
    --bronze: #cd7f32;
    --silver: #c0c0c0;
    --gold: #ffd700;

```

```
--platinum: #e5e4e2;
}

body {
  font-family: 'Inter', system-ui, -apple-system, sans-serif;
  background-color: var(--bg-primary);
  color: var(--text-primary);
  min-height: 100vh;
  display: flex;
  justify-content: center;
  align-items: flex-start;
  padding: 40px 20px;
  overflow-x: hidden;
  position: relative;
}

/* ===== Subtle Background Glow ===== */
.background-orbs {
  position: fixed;
  inset: 0;
  overflow: hidden;
  pointer-events: none;
  z-index: 0;
}

.orb {
  position: absolute;
  border-radius: 50%;
  filter: blur(120px);
  opacity: 0.08;
  background: #ffffff;
  animation: float 24s ease-in-out infinite;
}

.orb-1 {
  width: 500px;
  height: 500px;
  top: -10%;
  right: -10%;
}
```

```
.orb-2 {
  width: 400px;
  height: 400px;
  bottom: -5%;
  left: -5%;
  animation-delay: -8s;
}

.orb-3 {
  width: 300px;
  height: 300px;
  top: 40%;
  left: 50%;
  animation-delay: -16s;
}

@keyframes float {
  0%, 100% { transform: translate(0, 0) scale(1); }
  25% { transform: translate(40px, -30px) scale(1.05); }
  50% { transform: translate(-20px, 20px) scale(0.95); }
  75% { transform: translate(30px, 40px) scale(1.03); }
}

/* ===== Container ===== */
.container {
  position: relative;
  z-index: 1;
  width: 100%;
  max-width: 600px;
}

/* ===== Header ===== */
.header {
  text-align: center;
  margin-bottom: 32px;
}

.logo {
  display: flex;
```

```
    align-items: center;
    justify-content: center;
    gap: 14px;
    margin-bottom: 10px;
}

.logo-icon {
    display: flex;
    align-items: center;
    animation: bounce 2.5s ease-in-out infinite;
}

.logo-icon svg {
    width: 42px;
    height: 42px;
}

@keyframes bounce {
    0%, 100% { transform: translateY(0); }
    50% { transform: translateY(-5px); }
}

.header h1 {
    font-size: 2rem;
    font-weight: 800;
    color: var(--white);
    letter-spacing: -0.03em;
}

.subtitle {
    color: var(--text-secondary);
    font-size: 0.95rem;
    font-weight: 400;
}

/* ===== Main Card ===== */
.main-card {
    background: var(--bg-card);
    border: 1px solid var(--border-color);
    border-radius: 20px;
```

```
padding: 36px 32px;
backdrop-filter: blur(20px);
-webkit-backdrop-filter: blur(20px);
box-shadow: var(--shadow-glow);
margin-bottom: 28px;
}

/* ===== Input Section ===== */
.input-section {
  display: flex;
  flex-direction: column;
  gap: 16px;
}

.input-label {
  font-size: 0.8rem;
  font-weight: 600;
  text-transform: uppercase;
  letter-spacing: 0.1em;
  color: var(--text-secondary);
}

.input-wrapper {
  position: relative;
  display: flex;
  align-items: center;
}

.input-wrapper input {
  width: 100%;
  padding: 16px 80px 16px 20px;
  font-size: 1.6rem;
  font-weight: 700;
  font-family: 'Inter', sans-serif;
  color: var(--text-primary);
  background: rgba(255, 255, 255, 0.04);
  border: 2px solid var(--border-color);
  border-radius: 14px;
  outline: none;
  transition: border-color 0.3s ease, box-shadow 0.3s ease;
```

```

    -moz-appearance: textfield;
}

.input-wrapper input::-webkit-outer-spin-button,
.input-wrapper input::-webkit-inner-spin-button {
    -webkit-appearance: none;
    margin: 0;
}

.input-wrapper input::placeholder {
    color: rgba(255, 255, 255, 0.12);
    font-weight: 500;
}

.input-wrapper input:focus {
    border-color: rgba(255, 255, 255, 0.25);
    box-shadow: 0 0 0 4px rgba(255, 255, 255, 0.05);
}

.input-suffix {
    position: absolute;
    right: 20px;
    color: var(--text-secondary);
    font-size: 1rem;
    font-weight: 500;
    pointer-events: none;
}

/* ===== Button ===== */
#checkBtn {
    display: flex;
    align-items: center;
    justify-content: center;
    gap: 10px;
    width: 100%;
    padding: 16px;
    font-size: 1rem;
    font-weight: 600;
    font-family: 'Inter', sans-serif;
    color: #0e0e12;
}

```



```

    background: var(--white);
    border: none;
    border-radius: 14px;
    cursor: pointer;
    transition: transform 0.2s ease, box-shadow 0.2s ease, background 0.2s
ease;
}

#checkBtn:hover {
    transform: translateY(-2px);
    box-shadow: 0 8px 30px rgba(255, 255, 255, 0.12);
    background: #e8e8ed;
}

#checkBtn:active {
    transform: translateY(0);
}

/* ===== Result Section ===== */
.result-section {
    margin-top: 28px;
    display: none;
}

.result-section.visible {
    display: block;
    animation: slideUp 0.5s cubic-bezier(0.16, 1, 0.3, 1);
}

@keyframes slideUp {
    from { opacity: 0; transform: translateY(20px); }
    to { opacity: 1; transform: translateY(0); }
}

.result-card {
    text-align: center;
    padding: 32px 24px;
    border-radius: 16px;
    border: 1px solid var(--border-color);
    background: rgba(255, 255, 255, 0.02);

```

```
    position: relative;
    overflow: hidden;
}

.result-card::before {
    content: '';
    position: absolute;
    top: 0;
    left: 0;
    right: 0;
    height: 3px;
}

.result-card.not-eligible::before { background: var(--not-eligible); }
.result-card.bronze::before { background: var(--bronze); }
.result-card.silver::before { background: var(--silver); }
.result-card.gold::before { background: var(--gold); }
.result-card.platinum::before { background: var(--platinum); }

.result-icon {
    margin-bottom: 16px;
    display: flex;
    justify-content: center;
    animation: popIn 0.4s cubic-bezier(0.16, 1, 0.3, 1) 0.15s both;
}

.result-icon svg {
    width: 64px;
    height: 64px;
}

@keyframes popIn {
    from { opacity: 0; transform: scale(0.5); }
    to { opacity: 1; transform: scale(1); }
}

.result-title {
    font-size: 1.5rem;
    font-weight: 800;
    margin-bottom: 6px;
}
```

```
        letter-spacing: -0.01em;
    }

    .result-card.not-eligible .result-title { color: var(--not-eligible); }
    .result-card.bronze .result-title { color: var(--bronze); }
    .result-card.silver .result-title { color: var(--silver); }
    .result-card.gold .result-title { color: var(--gold); }
    .result-card.platinum .result-title { color: var(--platinum); }

    .result-marks {
        font-size: 0.9rem;
        color: var(--text-secondary);
        margin-bottom: 8px;
        font-weight: 500;
    }

    .result-message {
        font-size: 0.95rem;
        color: var(--text-secondary);
        line-height: 1.5;
    }

    /* ===== Criteria Section ===== */
    .criteria-section {
        margin-bottom: 40px;
    }

    .criteria-heading {
        font-size: 0.8rem;
        font-weight: 600;
        text-transform: uppercase;
        letter-spacing: 0.1em;
        color: var(--text-secondary);
        margin-bottom: 16px;
        text-align: center;
    }

    .criteria-grid {
        display: flex;
        flex-direction: column;
```

```
    gap: 8px;
}

.criteria-item {
    display: flex;
    align-items: center;
    gap: 14px;
    padding: 14px 20px;
    background: var(--bg-card);
    border: 1px solid var(--border-color);
    border-radius: 12px;
    backdrop-filter: blur(10px);
    transition: transform 0.2s ease, background 0.2s ease;
    position: relative;
    overflow: hidden;
}

.criteria-item:hover {
    transform: translateX(4px);
    background: var(--bg-card-hover);
}

.criteria-bar {
    position: absolute;
    left: 0;
    top: 0;
    bottom: 0;
    width: 3px;
}

.criteria-item.not-eligible .criteria-bar { background:
var(--not-eligible); }
.criteria-item.bronze .criteria-bar { background: var(--bronze); }
.criteria-item.silver .criteria-bar { background: var(--silver); }
.criteria-item.gold .criteria-bar { background: var(--gold); }
.criteria-item.platinum .criteria-bar { background: var(--platinum); }

.criteria-icon {
    display: flex;
    align-items: center;
```

```
        flex-shrink: 0;
    }

.criteria-icon svg {
    width: 24px;
    height: 24px;
}

.criteria-range {
    font-size: 0.9rem;
    font-weight: 600;
    color: var(--text-primary);
    min-width: 80px;
}

.criteria-label {
    font-size: 0.85rem;
    font-weight: 500;
}

.criteria-item.not-eligible .criteria-label { color: var(--not-eligible); }

.criteria-item.bronze .criteria-label { color: var(--bronze); }
.criteria-item.silver .criteria-label { color: var(--silver); }
.criteria-item.gold .criteria-label { color: var(--gold); }
.criteria-item.platinum .criteria-label { color: var(--platinum); }

/* ===== Shake Animation ===== */
@keyframes shake {
    0%, 100% { transform: translateX(0); }
    20% { transform: translateX(-8px); }
    40% { transform: translateX(8px); }
    60% { transform: translateX(-4px); }
    80% { transform: translateX(4px); }
}

.shake {
    animation: shake 0.4s ease;
}
```

```
/* ===== Responsive ===== */
@media (max-width: 480px) {
  body { padding: 24px 16px; }
  .header h1 { font-size: 1.5rem; }
  .main-card { padding: 28px 20px; border-radius: 16px; }
  .input-wrapper input { font-size: 1.3rem; padding: 14px 70px 14px
16px; }
  .result-icon svg { width: 48px; height: 48px; }
  .result-title { font-size: 1.3rem; }
}
```

Output:



Scholarship Eligibility

Enter your marks to discover your scholarship category

ENTER YOUR MARKS

90

/ 100

Check Eligibility →



Gold Scholarship

Your Marks: 90 / 100

Outstanding! You've earned a Gold Scholarship.

SCHOLARSHIP CRITERIA



Below 50

Not Eligible



50 – 69

Bronze



70 – 84

Silver



85 – 94

Gold



95 – 100

Platinum

Result/Conclusion:

The **Scholarship Eligibility Checker** was successfully developed using **HTML5, CSS3, and JavaScript**. The application accurately determines the scholarship category based on the marks entered by the user by applying conditional statements. It validates the input, provides immediate feedback, and dynamically displays the corresponding scholarship category with a modern and responsive interface. This case study demonstrates the practical implementation of HTML forms, CSS styling, JavaScript decision-making, and dynamic webpage interaction in a real-world educational application.

Case Study No.: 4

Case Study Title

ATM Card PIN Verification System Using JavaScript Functions

Theory

An **ATM Card PIN Verification System** is a simple security application that verifies whether the entered ATM PIN is a **palindrome**. A palindrome is a sequence of digits or characters that reads the same from left to right and right to left. Although real banking systems do not use palindrome checking for authentication, this case study demonstrates the practical implementation of JavaScript functions, scope, closures, and exception handling in a real-world scenario.

The application accepts a PIN from the user, reverses the entered PIN using a function, compares it with the original PIN, and displays whether it is a palindrome. If the PIN is a palindrome, a special security message is displayed; otherwise, the user is informed that the PIN is not a palindrome.

Function Declaration

A function declaration defines a reusable block of code using the **function** keyword. It is used to perform operations such as cleaning or validating the entered PIN.

Example:

- `function cleanPin(pin){`
- `return pin.trim();`
- `}`

Function Expression

A function expression stores a function inside a variable. It is commonly used for reversing the entered PIN.

Example:

- `const reversePin = function(pin){`
- `return pin.split('').reverse().join('');`
- `};`

Arrow Function

Arrow functions provide a shorter syntax for writing functions and are useful for simple comparisons.

Example:

```
const isPalindrome = (original, reversed) => original ===  
  reversed;
```

Scope

Scope defines where variables are accessible within a program.

- **Global Scope** – Variables declared outside functions are accessible throughout the program.
 - **Local Scope** – Variables declared inside functions can only be accessed within that function.
 - **Block Scope** – Variables declared using `let` and `const` inside a block `{ }` are limited to that block.
-

Closure

A **closure** is a function that remembers variables from its outer function even after the outer function has finished execution. In this case study, a closure is used to count the number of PIN verification attempts.

Working of the System

1. The user enters an ATM PIN.
2. JavaScript validates the entered PIN.
3. The PIN is reversed using a function expression.
4. An arrow function compares the original and reversed PIN.

5. If both values are equal, the PIN is identified as a palindrome.
 6. A special security message is displayed for palindrome PINs.
 7. A closure keeps track of the number of verification attempts.
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Features

- Accepts ATM PIN from the user.
 - Reverses the PIN using a JavaScript function.
 - Checks whether the PIN is a palindrome.
 - Demonstrates Function Declaration.
 - Demonstrates Function Expression.
 - Demonstrates Arrow Function.
 - Demonstrates Global and Local Scope.
 - Demonstrates Closure.
 - Displays user-friendly success and error messages.
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Applications

- Banking System Simulations
 - ATM Security Demonstrations
 - JavaScript Learning Projects
 - Programming Logic Development
 - Educational Applications
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Advantages

- Improves understanding of JavaScript functions.
- Demonstrates real-world implementation of function concepts.
- Enhances logical thinking.
- Provides hands-on experience with closures and scope.
- Interactive and user-friendly interface.
- Easy to understand and maintain.

1. Case Study Program Code:

index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>ATM Portal - Palindrome PIN Verification</title>

<style>

*{
    margin:0;
    padding:0;
    box-sizing:border-box;
    font-family:'Segoe UI',Tahoma,Geneva,Verdana,sans-serif;
}

body{
    height:100vh;
    display:flex;
    justify-content:center;
    align-items:center;
    background:linear-gradient(135deg,#0f172a,#1e3a8a,#3b82f6);
}

.atm-container{
    width:400px;
```

```
background:#fff;
border-radius:20px;
padding:35px;
box-shadow:0 20px 45px rgba(0,0,0,.25);
}

.header{
  text-align:center;
  margin-bottom:30px;
}

.user-badge{
  color:#64748b;
  text-transform:uppercase;
  letter-spacing:2px;
  font-size:12px;
}

.header h2{
  color:#2563eb;
  margin:10px 0;
}

.user-name{
  color:#111827;
  font-size:18px;
  font-weight:600;
  line-height:1.6;
}

.input-group{
  margin:25px 0;
}

label{
  display:block;
  margin-bottom:10px;
  color:#475569;
  font-size:13px;
  font-weight:600;
}
```

```
}

input{
    width:100%;
    padding:15px;
    border:2px solid #dbeafe;
    border-radius:12px;
    background:#f8fafc;
    font-size:22px;
    text-align:center;
    letter-spacing:5px;
    transition:.3s;
}

input:focus{
    outline:none;
    border-color:#2563eb;
    box-shadow:0 0 12px rgba(37,99,235,.25);
}

button{
    width:100%;
    padding:15px;
    border:none;
    border-radius:12px;
    background:linear-gradient(135deg,#2563eb,#1d4ed8);
    color:white;
    font-size:16px;
    font-weight:bold;
    cursor:pointer;
    transition:.3s;
}

button:hover{
    transform:translateY(-2px);
    box-shadow:0 12px 25px rgba(37,99,235,.35);
}

.screen-display{
    margin-top:25px;
}
```

```
    min-height:80px;
    padding:18px;
    border-radius:12px;
    display:flex;
    justify-content:center;
    align-items:center;
    text-align:center;
    background:#f8fafc;
    border:2px solid #e2e8f0;
    color:#64748b;
    line-height:1.5;
    transition:.3s;
}

.special-msg{
    background:#dcfce7;
    color:#166534;
    border-color:#22c55e;
    font-weight:bold;
}

.error-msg{
    background:#fee2e2;
    color:#b91c1c;
    border-color:#ef4444;
    font-weight:bold;
}

.footer{
    margin-top:20px;
    text-align:center;
    color:#94a3b8;
    font-size:12px;
}

</style>
</head>

<body>
```



```
<div class="atm-container">

  <div class="header">

    <div class="user-badge">
    Secure ATM Portal
    </div>

    <h2><img alt="ATM icon" data-bbox="162 255 180 268" /> ATM LOGIN</h2>

    <div class="user-name">
    Radhika Chaudhari<br>
    PRN : 24070521150
    </div>

  </div>

  <div class="input-group">

    <label>ENTER YOUR 4-DIGIT PIN</label>

    <input
    type="text"
    id="pinInput"
    maxlength="4"
    placeholder="Enter PIN"
    autocomplete="off"
    inputmode="numeric">

  </div>

  <button id="verifyBtn">
  Verify PIN
  </button>

  <div class="screen-display" id="atmScreen">
  Welcome! Please enter your 4-digit PIN.
  </div>

  <div class="footer">
```

```
Palindrome PIN Verification System
</div>

</div>

<script>

// Function Declaration
function reversePIN(pin) {
    return pin.split('').reverse().join('');
}

// Function Expression
const isPalindrome = function(pin) {
    return pin === reversePIN(pin);
};

// Closure
function createATMVerifier(){

    const successMessage =
        "✅ PIN Verified Successfully! Welcome to Secure ATM Banking.";

    // Arrow Function
    const verifyATMCard = (enteredPin)=>{

        if(enteredPin.length !== 4 || isNaN(enteredPin)){

            return{
                status:"error",
                message:"❌ Please enter a valid 4-digit numerical PIN."
            };

        }

        if(isPalindrome(enteredPin)){

            return{
                status:"special",
                message:successMessage
            };
        }
    };
}
```

```

        };

    }

    return{

        status:"error",
        message:"❌ Access Denied! PIN is NOT a Palindrome."

    };

};

return verifyATMCard;
}

const atmSystem = createATMVerifier();

const pinInput = document.getElementById("pinInput");
const verifyBtn = document.getElementById("verifyBtn");
const atmScreen = document.getElementById("atmScreen");

// Allow only digits
pinInput.addEventListener("input", ()=>{

    pinInput.value = pinInput.value.replace(/[^0-9]/g, "");

});

// Verify Button
verifyBtn.addEventListener("click", ()=>{

    const result = atmSystem(pinInput.value);

    atmScreen.className = "screen-display";

    if(result.status==="special"){

        atmScreen.classList.add("special-msg");
    }

```

```
        alert(
"⚠ SECURITY WARNING!\n\n" +
"The PIN you entered is a PALINDROME.\n\n" +
"Palindrome PINs are considered weak because they are easier to
guess.\n\n" +
"Please change your PIN after logging in for better security."
        );

    }

    else{

        atmScreen.classList.add("error-msg");

    }

    atmScreen.innerHTML = result.message;

});

// Press Enter to Verify
pinInput.addEventListener("keypress", (e)=>{

    if(e.key==="Enter"){

        verifyBtn.click();

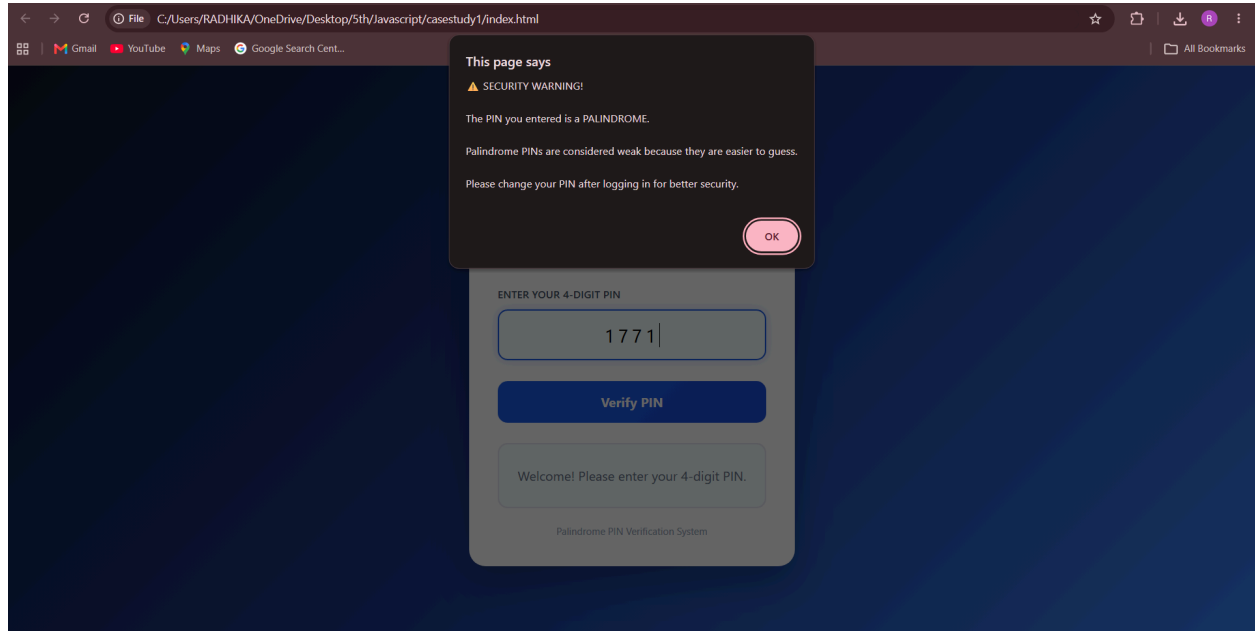
    }

});

</script>

</body>
</html>
```

Output:



Result/Conclusion:

The **ATM Card PIN Verification System** was successfully implemented using **HTML5, CSS3, and JavaScript**. The application accepts an ATM PIN, reverses it using a function expression, and checks whether it is a palindrome using an arrow function. The project demonstrates the concepts of **function declaration, function expression, arrow functions, variable scope, closures, and JavaScript-based validation**. The system accurately identifies palindrome and non-palindrome PINs while providing an interactive and user-friendly interface, thereby enhancing the understanding of JavaScript functions and their real-world applications.

